T3 group factors: EFT_1_after_S1

Threshold-order-aware EFT stage. Active heavy-field content: F, S2.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1,\ m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2}\lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HHS_1S_2^\dagger + \mathrm{h.c.}$
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-B	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	S1	F, S2

Active renormalisable T3 quantities

g_1, g_2, g_3, Y_u, Y_d, Y_e and the SM Higgs couplings remain active at every stage.

quantity	T3 − B ($\alpha = -1$)
y_1	–
y_2	✓
M_F	✓
m_1^2	–
m_2^2	✓
$\lambda_{S_1}^{(1)}$	–
$\lambda_{S_2}^{(1)}$	✓
$\lambda_{HS_1}^{(1)}$, $\lambda_{HS_1}^{(A)}$	–
$\lambda_{HS_2}^{(1)}$, $\lambda_{HS_2}^{(A)}$	✓
λ_5, $\lambda_{12}^{(1)}$, $\lambda_{12}^{(A)}$, $\lambda_{12}^{(\times)}$	–

Stage-specific group-factor status

The threshold ordering at this stage is supported by the report layer, so a PDF is produced for it. A dedicated analytic renormalisable-RGE group-factor module is not currently available for this reduced heavy-field content. Coefficients from the UV theory or from the special F-first EFT are therefore not reused here, because doing so would assign beta-function coefficients from the wrong dynamical theory.

Representation invariants

Field representations

run	S_1	S_2	F	α
$\mathrm{T3-B}$	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	-1

Next threshold

The next GroupFactors PDF is generated from the next serialized EFT-stage label in the threshold plan, independent of which field is removed next.